

Deepa L. Ramamurthy, PhD

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RESEARCH INTERESTS

I am a systems neuroscientist interested in the interplay between cognition and perception. My research program combines quantitative behavioral methods with cell type-specific tools to probe neural circuit function, with the goal of understanding how experience dynamically shapes sensory representations across timescales to guide adaptive behavior.

POSITIONS & EMPLOYMENT

Assistant Professor , Psychology Department, UC Riverside	Aug. 2025-present
Postdoctoral Fellow / Assistant Project Scientist , UC Berkeley Advisor: Dr. Daniel E. Feldman	2018 - 2025
Graduate Student , UC Davis Advisor: Dr. Leah A. Krubitzer	2012 - 2018
Undergraduate Honors / Post-Bac Researcher , Stanford University Advisor: Dr. Eric I. Knudsen	2010 - 2012
Student Intern , Sleep Research Lab, SRI International, Menlo Park, CA	2011

EDUCATION

PhD, Neuroscience , UC Davis, Davis, CA	Sept. 2018
BS, Biology with Honors , <i>summa cum laude</i> , Notre Dame de Namur University, Belmont, CA	May 2011

ADDITIONAL RESEARCH TRAINING

CCN 2022 Workshop on Calcium & Voltage Imaging Analysis, Flatiron Institute, New York, NY	2022
Summer Workshop on the Dynamic Brain, Allen Institute for Brain Science & UW, Seattle, WA	2018
Neural Systems & Behavior Course, Marine Biological Laboratory, Woods Hole, MA	2017
Cognition, Brain and Technology Summer School, Universitat Pompeu Fabra, Barcelona, Spain	2014

GRANTS & FELLOWSHIPS

BBRF Young Investigator Grant "Defining the roles of cortical gatekeeper cells in maladaptive attention"	2027 - 2029
Regents Faculty Fellowship	2026 - 2027
K99/R00 NIH BRAIN Initiative Advanced Postdoctoral Career Transition Award "Cell type-specific mechanisms of history-dependent perceptual biases in sensory cortex"	2023 - 2028
Ruth L. Kirschstein National Research Service Award "VIP interneuron circuits as a mechanism for attentional selection in mouse somatosensory cortex"	2020 - 2022
National Science Foundation Graduate Research Fellowship "Age-dependence of compensatory cross-modal plasticity in the mammalian neocortex"	2014 - 2017

AWARDS & HONORS

Simons Foundation Center for Computational Neuroscience Travel Grant	2022
Leading Edge Fellow	2020
Society for Neuroscience Trainee Professional Development Award	2020
Ling-Lie Chau Graduate Student Award for Brain Research, UC Davis	2017
Grass Foundation Neural Systems & Behavior Course Scholarship Award	2017

UC Davis Graduate Student Association Spring Travel Award	2015
UC Davis Center for Vision Science Travel Fellowship Award	2014
UC Davis NEI Vision Science Training Grant (NIH T32) Predoctoral Award	2013
Outstanding Academic Achievement Award in Biology (NDNU Dept. of Natural Sciences)	2011
Palo Alto Medical Foundation Science Scholarship	2009

PUBLICATIONS

* undergraduate mentee co-authors, ^ corresponding author

Google Scholar Profile: <https://scholar.google.com/citations?user=rx0-fg8AAAAJ&hl=en>

Ramamurthy DL[^], Englund M, Kovacs TJ*, Dodson H*, Krubitzer LA. (2026). Complex three-dimensional rearing environments amplify compensatory plasticity following early blindness. *eNeuro* 13 (7) ENEURO.0059-26.2026. doi: <https://doi.org/10.1523/ENEURO.0059-26.2026>. **[Featured as a Research Spotlight article]**

Ramamurthy DL, Rodriguez L, Cen C*, Li S*, Chen A*, Feldman DE[^]. (2025). Reward history guides focal attention in whisker somatosensory cortex. *Nat. Commun* 16, 5580. doi: <https://doi.org/10.1038/s41467-025-60592-w>.

Ramamurthy DL[^], Chen A*, Zhou J*, Park C*, Huang PC*, Bhargavan P*, Krishna G*, Liu J*, Casale K*, Feldman DE[^]. (2023). VIP interneurons in sensory cortex encode sensory and action signals but not direct reward signals. *Curr Biol* 33 (16): 3398-3408.E7. doi: <https://doi.org/10.1016/j.cub.2023.06.086>. **[Featured in Current Biology Dispatch: <https://doi.org/10.1016/j.cub.2023.07.059>]**

Meleppat RK, Fortenbach CR, Jian Y, Martinez ES, Wagner K, Modjtahedi BS, Motta MJ, **Ramamurthy DL**, Schwab IR, and Zawadzki RJ[^] (2022) In vivo imaging of retinal and choroidal morphology and vascular plexuses of vertebrates using swept-source optical coherence tomography. *Transl Vis Sci Technol* 11(8):11. doi: <https://doi.org/10.1167/tvst.11.8.11>.

Ramamurthy DL, Dodson HK*, Krubitzer LA[^] (2021) Developmental plasticity of texture discrimination following early vision loss in the marsupial *Monodelphis domestica*. *J Exp Biol* 224 (9): jeb236646. doi: <https://doi.org/10.1242/jeb.236646>.

Ramamurthy DL[^], Recanzone GH (2020) Age-related changes in intensity coding of onset and offset responses in auditory cortical fields A1 and CL of rhesus macaques. *J Neurophysiol.* 123(3):1015-1025. doi: <https://doi.org/10.1152/jn.00373.2019>. **[American Physiological Society APSselect award article: <https://journals.physiology.org/apsselect/archive>]**

Ramamurthy DL, Krubitzer LA[^] (2018) Neural coding of whisker-mediated touch in primary somatosensory cortex is altered following early blindness. *J Neurosci.* 38 (27) 6172-6189. doi: <https://doi.org/10.1523/JNEUROSCI.0066-18.2018>.

Ramamurthy DL[^], Recanzone GH (2017) Spectral and spatial tuning of onset and offset response functions in auditory cortical fields A1 and CL of rhesus macaques. *J Neurophysiol.* 117(3):966-986. doi: <https://doi.org/10.1152/jn.00534.2016>. **[H1 Connect/Faculty of 1000 recommended article: <https://connect.h1.co/article/727086333>]**

Ramamurthy DL, Krubitzer LA[^] (2016) The evolution of whisker-mediated somatosensation in mammals: sensory processing in barrelless S1 cortex of a marsupial, *Monodelphis domestica*. *J Comp Neurol* 524(17):3587-3613. doi: <https://doi.org/10.1002/cne.24018>.

Sridharan D[^], **Ramamurthy DL**, Schwarz JS, Knudsen EI (2014) Visuospatial selective attention in chickens. *PNAS* 111(19):E2056-65. doi: <https://doi.org/10.1073/pnas.1316824111>.

Sridharan D[^], **Ramamurthy DL**, Knudsen EI (2013) Spatial probability dynamically modulates visual target detection in chickens. *PLOS ONE* 8(5): e64136. doi: <https://doi.org/10.1371/journal.pone.0064136>.

INVITED TALKS

UCR Neuroscience Seminar Series, University of California, Riverside, Riverside, CA	Oct. 2025
Mount Sinai Neuroscience seminars, Icahn School of Medicine at Mount Sinai, New York, NY	June 2025
Department of Neurobiology and Biophysics, University of Washington, Seattle, WA	March 2025

Department of Psychology, University of California, Riverside, Riverside, CA	Feb. 2025
Department of Zoology and Physiology, University of Wyoming, Laramie, WY	Feb. 2025
Department of Biology, Texas A&M University, College Station, TX	Jan. 2025
SYNAPSES seminar at Yale Neuroscience, Yale University	Dec. 2024
Leading Edge Retreat, Cold Spring Harbor Laboratory	Aug. 2024
Neurotech@Berkeley Research Panel, UC Berkeley, Berkeley, CA	Feb. 2024
Barrels XXXV, 35th Annual Barrels Meeting, La Jolla, CA	Nov. 2022
Leading Edge Symposium, HHMI/Janelia Research Campus Conference	June 2021
William Woods University, Fulton, MO	Apr. 2021
Nanosymposium, "Touch, Itch and Pain," Society for Neuroscience Meeting, Washington, DC	Nov. 2017

CONFERENCE PRESENTATIONS

** undergraduate mentee co-authors*

- Ramamurthy DL.** (2026). Dissociable effects of selective attention and global arousal in sensory cortex. Presentation. Neural Circuits in Health and Disease. 2026 Summer Conference, UC Irvine Center for Neural Circuit Mapping (CNCM). August 2026. Irvine, CA. **[CNCM Poster Award]**
- Ramamurthy DL.** (2026). Dissociable effects of selective attention and global arousal on neuronal populations in sensory cortex. Presentation. 2026 NIH BRAIN Initiative Conference. August 2026. Rockville, MD.
- Cen C*, Li S*, Rodriguez L, Feldman DE, **Ramamurthy DL.** Dissociable effects of selective attention and global arousal in whisker somatosensory cortex. Program No. 119.07. 2025 Neuroscience Meeting Planner. San Diego: Society for Neuroscience, 2025.
- Ramamurthy DL,** Rodriguez L, Cen C*, Li S*, Chen A*, Feldman DE. Attentional capture by reward history in whisker somatosensory cortex. Program No. 340.24. 2024 Neuroscience Meeting Planner. Chicago, IL: Society for Neuroscience, 2024.
- Ramamurthy DL,** Rodriguez L, Cen C*, Li S*, Chen A*, Feldman DE. Reward history guides attentional selection in mouse sensory cortex. 10th Annual BRAIN Initiative Conference. NIH BRAIN Initiative Conference. Rockville, MD, 2024.
- Ramamurthy DL,** Chen A*, Feldman DE. Neural correlates of history-based selective attention in mouse primary somatosensory cortex. Neurobiology of Cognition Gordon Research Conference: Neural Circuits, Dynamics and Computations of Cognition and Behavior. Newry, ME, 2024.
- Ramamurthy DL,** Chen A*, Feldman DE. Neural correlates of history-based selective attention in mouse primary somatosensory cortex. Program No. 292.03. 2022 Neuroscience Meeting Planner. San Diego, CA: Society for Neuroscience, 2022.
- Ramamurthy DL,** Chen A*, Huang PC*, Bharghavan P*, Krishna G*, Casale K*, Feldman DE. Sensory and action-related responses of VIP interneurons in mouse whisker S1 during goal-directed behavior. Program No. P412.03. 2021 Neuroscience Meeting Planner. Virtual: Society for Neuroscience, 2021.
- Ramamurthy DL,** Chen A*, Krishna G*, Bharghavan P*, Casale K*, Feldman DE. Recent trial history cues attentional selection of whisker stimuli in mice. SfN Global Connectome: A Virtual Event. 2021. Digital Abstract. Society for Neuroscience. Virtual Conference. **[Received Society for Neuroscience 2020 Trainee Professional Development Award]**
- Ramamurthy DL,** Feldman DE. Recent trial history cues attentional selection of whisker stimuli in mice. BARRELS XXXIII. Poster. 33rd Annual Barrels Society Meeting, 2020. SfN Satellite Event. Virtual Conference.
- Ramamurthy DL,** Englund MC, Krubitzer LA. Alterations in somatosensory receptive fields following early blindness are shaped by the environment. Program No. 764.02. 2018 Neuroscience Meeting Planner. San Diego, CA: Society for Neuroscience, 2018.
- Bernardez S, **Ramamurthy D,** Browning Y. Influence of expectation and experience on sensory coding during a visual change detection task. Student team project. Summer Workshop on the Dynamic Brain. Allen Institute for Brain Science. Seattle, Washington. 2018. **[Awarded 'Best in Show']**
- Ramamurthy DL,** Krubitzer LA. Effect of environmental enrichment on somatosensory plasticity in the

neocortex following early blindness. Program No. 15.09. 2017 Neuroscience Meeting Planner. Washington, DC: Society for Neuroscience, 2017. **[Nanosymposium talk. Highlighted as a Neuroscience 2017 Hot Topic presentation]**

Ramamurthy DL, Krubitzer LA. Effect of environmental enrichment on somatosensory plasticity in the neocortex following early blindness. BARRELS XXX. Poster. 30th Annual Barrels Society Meeting, 2017. SfN Satellite Event. Baltimore, MD.

Ramamurthy DL, Krubitzer LA. Receptive fields and response properties of neurons in the S1 whisker representation of early blind short-tailed opossums. Program No. 709.06. 2016 Neuroscience Meeting Planner. San Diego, CA: Society for Neuroscience, 2016.

Foreman K, **Ramamurthy D**, Weller C, Krubitzer LA, Stolzenberg DS. Genetic and epigenetic regulation of the cortical phenotype: The effects of early bilateral enucleation on epigenetic and genetic modifications in developing neocortex. Program No. 678.06. 2016 Neuroscience Meeting Planner. San Diego, CA: Society for Neuroscience, 2016.

Ramamurthy DL, Krubitzer LA. Receptive fields and response characteristics of neurons in the S1 whisker representation of the short-tailed opossum, *Monodelphis domestica*. Society for Neuroscience Annual Meeting. Program No. 516.06. 2015 Neuroscience Meeting Planner. Chicago, IL: Society for Neuroscience, 2015.

Ramamurthy DL, Krubitzer LA. Somatosensory processing in barrelless S1 cortex of a marsupial, *Monodelphis domestica*. 2015 JBJC Meeting. Datablitz. JB Johnston Club for Evolutionary Neuroscience Annual Meeting, 2015. SfN Satellite Event. Chicago, IL.

Tardio C, **Ramamurthy D**, Camboni D, Maffei G, Glancy J, Manoliu L, Sanchez M, Martinez PA, Brandi S, Rongala U (2014) Counterbalancing perturbations in a self-balancing robot through a cerebellar learning model. Student team project. Barcelona Cognition, Brain and Technology (BCBT) Summer School. **[Awarded 2nd place]**

Ramamurthy DL, Gordon AG*, Krubitzer LA. Functional topography and tuning properties of neurons in the S1 whisker representation of the short-tailed opossum, *Monodelphis domestica*. Program No. 440.14. 2014 Neuroscience Meeting Planner. Washington, DC: Society for Neuroscience, 2014.

Ramamurthy DL, Ng C-W, Gray DT, Overton JA, Recanzone GH. Spectral and spatial tuning of onset and offset responses in the auditory cortex of awake macaque monkeys. Program No. 354.11. 2013 Neuroscience Meeting Planner. San Diego, CA: Society for Neuroscience, 2013.

Ramamurthy DL, Ng C-W, Gray DT, Overton JA, Recanzone GH. Spectral and spatial tuning of onset and offset responses in the auditory cortex of awake macaque monkeys. APAN XI: Program No. 60. Poster. Tucker-Davis Symposium on Advances and Perspectives in Auditory Neuroscience (APAN), 2013. SfN Satellite Event. San Diego, CA.

DEPARTMENTAL SEMINARS

UC Riverside NRSC Graduate Program Recruitment Talk	Oct. 2025
UC Riverside NRSC Graduate Program Pizza Lunch	Oct. 2025
UC Berkeley MCB Postdoc Research Showcase, Berkeley, CA	Mar. 2024, Mar. 2022
UC Berkeley HWNI Cortex Club, Berkeley, CA	Dec. 2020
UC Berkeley MCB Neurodinner, Berkeley, CA	Sept. 2019
UC Davis Neurolunch, Davis, CA	Mar. 2018
UC Davis Center for Vision Science Symposium, Davis, CA	Jan. 2018
UC Davis Annual Neuroscience Retreat, Tahoe City, CA	Sept. 2016, Sept. 2014

MENTORING

Graduate Students

Brontë Neal (PSYC), incoming, Fall 2026

Levi-Briana Shinn (NRSC), incoming rotation student, Fall 2026

Undergraduate Students**UC Riverside (2025 - present)**

Current. Mahi Ekmekci (UC LEADS & Honors Capstone), Sreeya Gambhirrao (Honors Capstone), Sarai Biemas, Emma Cohen, Yashica Pandey

Alumni. Sakshi Bhargava (UCR Undergraduate Research & Creative Activities Symposium), Katelyn Lau (UCR Undergraduate Research & Creative Activities Symposium), Debdeep Bandyopadhyay, Natalie Cortez, Avik Shrestha

UC Berkeley (2018 - 2025)

Celine Cen (URAP, MCB PUMP & Honors Research), Siqian Li (MCB PUMP & Honors Research), Chanbin Park (URAP), Jinjian Liu (URAP), Andrew Chen (SURF), Patrick Huang, Jiayu Zhou, Priyanka Bharghavan, Gayathri Krishna, Kayla Casale

UC Davis (2012 - 2018)

Krishna Yadav-Rajan (WiSE), Jessica (Sibal) Wheeler (NDNU alumna), Tanner Kovacs, Heather Dodson, Ian Strieter, Adam Gordon

Doctoral Committees

Luiza Rocha (NRSC, Zagha Lab, 2026 - present), Alexandra Tejada (NRSC, Zagha Lab, 2026 - present), Sarah Kabbara (NRSC, Zagha Lab, 2026 - present)

TEACHING**Courses**

Graduate Seminar in Behavioral Neuroscience: Neural Circuits for Attention (PSYC 263, UC Riverside, forthcoming; maximum enrollment of 15 students)	Fall 2026
Introductory Psychology (PSYC 001, UC Riverside; 568 students)	Winter 2026
Topics in Cognitive Neuroscience: Neural Plasticity (PSYC 136, UC Riverside; 23 students)	Fall 2025

Guest Lectures

Graduate Seminar in Professional Development (PSYC 409B, UC Riverside)	Spring 2026
Somatosensation 2: Pain, Itch and Temperature; Circuit, Systems & Behavioral Neuroscience (MCB 161, UC Berkeley)	Spring 2024
Neuroscience Research Seminar (NEUR 150, Santa Clara University)	Spring 2023

Additional Teaching Experience

Comparative Neuroanatomy (NPB/PSC 124, UC Davis)	Winter 2018
Comparative Neuroanatomy (NPB/PSC 124, UC Davis)	Winter 2015
Sanskrit Literature (SN/FC/KK32, Stella Maris College, Chennai, India)	Fall 2008

PROFESSIONAL DEVELOPMENT

Advancement and Retention in Academe (ARA) Program Workshop Series	Winter 2026
Teaching Foundations Workshop Series, Academy of Distinguished Teaching, UC Riverside	Fall 2025
Torrey Pines Academic Lab Management & Leadership Symposium	Spring 2021
Developed Circuit/Systems Neuroscience course materials in MCB 290 Neuroscience Coursework Design (Instructor: Dan Feldman)	Fall 2019

SERVICE

Facilitator, Build Your Brand Workshop, Leading Edge Program	Fall 2026, forthcoming
UCR Department of Psychology Undergraduate Education Committee	Fall 2026, forthcoming
UC LEADS Faculty Mentor	Summer 2026
Member, Faculty for Undergraduate Neuroscience	Spring 2026 - present
UCR Committee-in-Charge of the Neuroscience Undergraduate Program	Spring 2026 - present
UCR Undergraduate Research Journal Faculty Advisory Board	Spring 2026 - present

Abstract Reviewer, UCR R'PSYC Research Conference	Spring 2026
UCR Neuroscience Seminar Series Host	Spring 2026
UCR Minigrant Review Committee	Winter 2026 - present
Member, Society for Neuroscience	2012 - present
UCR Department of Psychology Diversity, Equity & Inclusion Committee	2025 - 2026
UCR NRSC booth at SfN Graduate School Fair, Faculty Representative	Fall 2025
Session Chair, Systems Neuroscience, Leading Edge Fellows Symposium	2022
Panelist, Navigating your Postdoc Panel, Leading Edge Program	2021
Student Member, J.B. Johnston Club for Evolutionary Neuroscience	2013 - 2018
Co-organizer, Student Organized Seminar Series, UC Davis Neuroscience Graduate Program	2013 - 2015

Ad hoc Reviewer

Nature Communications, Journal of Experimental Biology, Neuron, Journal of Comparative Physiology A

OUTREACH

Letters to a Pre-Scientist (https://prescientist.org/), STEM education pen pal program for middle school students in low-income schools	2015 - present
Brain Awareness Week outreach visit to Norco Community College, Norco, CA	2026
Project SHORT (https://www.project-short.com/), pro-bono mentoring for graduate school applicants from underrepresented populations in STEM	2020 - 2025
Research Presentation and Discussion Panel speaker, Neuroscience Initiative to Enhance Diversity (NIED), UC Davis Neuroscience Graduate Program	2015, 2016, 2017
UC Davis Center for Neuroscience Public Seminars	2016, 2017
Davis Science Collective outreach event at the Mary L. Stephens Davis Branch Library	2015, 2017
Krubitzer Lab outreach visit to California Academy of Sciences Brilliant! Science: Brain Power Family Festival.	2014
Brain Awareness Week outreach visit to the Met Sacramento High School	2014
Brain Awareness Week outreach visit to William Land Elementary School	2013